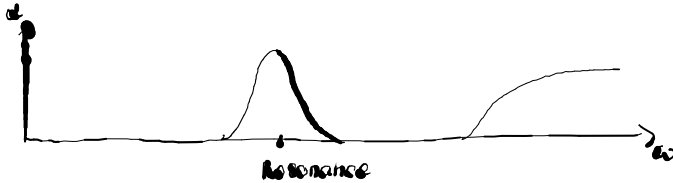


Why should we do solid states?

In nature, we usually observe linear absorption spectrum

$I_0 \rightarrow$ [material] $\rightarrow I_t \rightarrow$ $I_t = I_0 e^{-\alpha(\omega)L}$ $\alpha(\omega) = \frac{1}{L} \ln \left(\frac{I_0}{I_t} \right)$



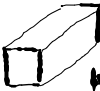
We can modify the absorption spectrum with a voltage. The electric field changes exciton's peak



Some modify with ultrafast laser



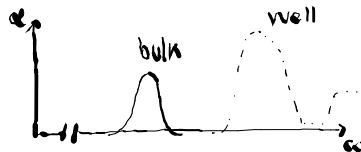
Some modify with quantum confinement


 bulk: a very small nano scale can be considered bulk.
 bulk has many electrons $\rightarrow$ band
 electron's wavelength is not confined


 Quantum well
 Electron is confined vertically
 ~100 Å

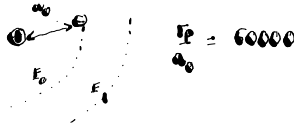

 Quantum wire

 Quantum dot
 Wire confines in two dimensions
 Dot confines in three dimensions

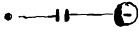


By Confinement, what is confined? The answer is interesting.

Consider a Hydrogen atom, Bohr radius is $a_0 = 0.529 \text{ nm} = 0.529 \text{ \AA}$. Meanwhile, proton has radius $r_p \approx 0.824 \text{ fm}$



Atom is mostly empty



Wave function $\psi(x)$



In solid, the distance between atom is $\sim 5 \text{ nm}$. Even when people say electron is "small", yes it is small in mass, but its probability spreads over $\sim \frac{1}{100} \text{ nm}$



Electron does not have a single momentum, it has a bandwidth

$$\lambda = \frac{h}{p} \Rightarrow \text{There is a band width in wavelength}$$

$$p = \hbar k \quad v = 10^6 \text{ m/s} \quad \lambda = 0.73 \text{ nm} \Rightarrow \text{This is roughly over } 10 \text{ atoms in a lattice}$$

$$\vec{p} = \hbar \vec{k} \quad \vec{v} = \frac{1}{m} \hbar \vec{k}$$

An electron can have nearly one momentum: small bandwidth. This is solid state



An electron can have a large momentum bandwidth



The bandwidth creates an envelope. This is why, we see



Somewhere in middle

$$\Delta k, n \quad \psi_1 + \psi_2 + \dots = \psi_n \quad \text{localized}$$

$$\Delta k \uparrow \quad \psi \rightarrow \delta$$

When we say confinement, what happens is actually allowing many k or λ where

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad \lambda_n = \frac{2L}{n} \quad \lambda_1 + \lambda_2 + \dots = \lambda$$

The superposition is confined. The electron is in a superposition of different possible momenta!

Note

$k, \Delta k$ electron $\neq$ a superposition electron of different k .

Cubic Basis

Consider the lattice, solid basically mean atoms or ions in periodic arrangement.

Ideal lattice positions follow formula

$$\vec{R}_n = n_1 \vec{a} + n_2 \vec{b} + n_3 \vec{c}$$

where n are integers and $\vec{a}$, $\vec{b}$, and $\vec{c}$ are primitive vectors.

Primitive means $\Omega_c = \vec{a} \cdot (\vec{b} \times \vec{c}) = \text{volume is minimized}$

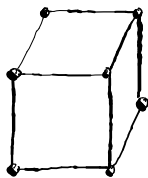
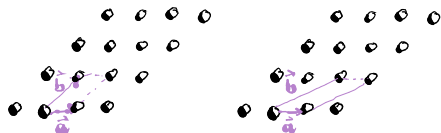
Primitives are not unique

Wigner-Seitz Primitive cell

Primitive vectors create the lattice not the cell

The primitive cell needs:

fills all space with translation
no gaps/overlaps, contains one exactly lattice point

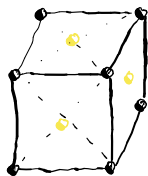


Each corner is shared with 8 cubes

$$N = 8 \cdot \frac{1}{8} = 1$$

The cell is primitive a^3

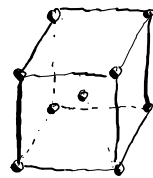
Simple Cubic



Face centered cubic

$$N = 8\left(\frac{1}{8}\right) + 6\left(\frac{1}{2}\right) = 4$$

The primitive cell is $\frac{a^3}{4}$



Body centered cubic

$N = 8\left(\frac{1}{8}\right) + 1 = 2$
The cell is primitive as $\frac{a^3}{2}$

Basis each lattice point has a basis which might be multiple atoms per point or complex molecules

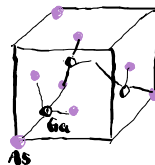


Crystal Structure = Lattice + Basis $\vec{r}_{n,\alpha} = \vec{R}_n + \vec{r}$ $\alpha = 1, 2, 3$

GaAs: zincblende $\vec{r} = \frac{a}{4}(1, 1, 1)$

FCC $\vec{r} = (0, 0, 0)$ $\vec{r} = \frac{a}{4}(1, 1, 1)$

This two FCC means GaAs can be pictured as two identical FCC slightly shifted



Reciprocal Lattice

Let $\rho(\vec{r})$ be electronic charge density at position $\vec{r}$, a lattice translation

$$\rho(\vec{r} + \vec{R}_n) = \rho(\vec{r}) \propto |\psi(\vec{r})|^2$$

This density should be identical by symmetry.

Write $\rho(\vec{r}) = \sum_{\vec{G}} S(\vec{G}) e^{i\vec{G} \cdot \vec{r}}$ which is simply sum of plane waves

$$\rho(\vec{r}) = \sum_{\vec{G}} S(\vec{G}) e^{i\vec{G} \cdot \vec{r}} \quad \vec{G} \text{ is spatial frequency } \lambda = \frac{2\pi}{|\vec{G}|} \quad S(\vec{G}) = \frac{1}{\Omega} \int \rho(\vec{r}) e^{-i\vec{G} \cdot \vec{r}} d^3r$$

A lattice translation says

$$\rho(\vec{r} + \vec{R}) = \sum_{\vec{G}} S(\vec{G}) e^{i\vec{G} \cdot (\vec{r} + \vec{R})} \Rightarrow \vec{G} \cdot \vec{R} = 2\pi N$$

$$\vec{a} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3 \quad \vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3$$

$$\vec{A} \cdot \vec{a} = \vec{B} \cdot \vec{b} = \vec{C} \cdot \vec{c} = 2\pi$$

This says only certain electron momentum or wavevector are allowed to preserve the state.

To find $\vec{A}, \vec{B}, \vec{C}$,

$$\vec{A} = \vec{a} \cdot (\vec{b} \times \vec{c}) \quad \vec{B} = \frac{2\pi}{\Omega} (\vec{c} \times \vec{a}) \quad \vec{C} = \frac{2\pi}{\Omega} (\vec{a} \times \vec{b})$$

If the primitive cell in reciprocal lattice looks like Wigner Seitz, it is Brillouin zone